ESP32 OTA Dashboard

Command Cheatsheet & Setup Guide

PC Server IP (LAN)	192.168.12.214
WSL IP (Internal)	172.28.135.37
Backend Port	8000
Frontend Port	5173
Backend URL	http://192.168.12.214:8000
Frontend URL (User)	http://192.168.12.214:5173
GitHub User	irvandimas840-art
Repo Backend	github.com/irvandimas840-art/local-server
Repo Frontend	github.com/irvandimas840-art/esp32-ota

1. Masuk ke WSL Ubuntu (dari PC)

Buka CMD atau PowerShell di PC, lalu ketik:

```
wsl -d Ubuntu-22.04
```

Untuk keluar dari WSL:

```
exit
```

2. Jalankan Backend (PC/WSL)

Masuk ke folder local-server dan jalankan uvicorn:

```
cd ~/local-server python3 -m uvicorn main:app --host 0.0.0.0 --port 8000 --reload
```

Test backend jalan — buka browser:

```
http://192.168.12.214:8000 http://192.168.12.214:8000/ota
```

* Harus muncul: {"status":"ok","message":"Local Server API running"}

3. Jalankan Frontend (PC/WSL)

Buka terminal WSL baru, masuk ke folder frontend:

```
cd /mnt/c/users/itd33/esp32-ota/frontend npm run dev -- --host
```

User di LAN bisa akses dashboard di:

```
http://192.168.12.214:5173
```

* Jalankan backend dan frontend di 2 terminal terpisah!

4. Port Proxy Windows ke WSL (PC)

Jalankan di CMD Windows sebagai Administrator:

```
# Port proxy backend netsh interface portproxy add v4tov4 listenport=8000
listenaddress=0.0.0.0 connectport=8000 connectaddress=172.28.135.37 # Port proxy frontend
netsh interface portproxy add v4tov4 listenport=5173 listenaddress=0.0.0.0 connectport=5173
connectaddress=172.28.135.37 # Firewall rule netsh advfirewall firewall add rule
name="uvicorn 8000" dir=in action=allow protocol=TCP localport=8000
```

* Lakukan sekali saja, berlaku sampai Windows restart.

5. Git — Laptop (Update Kode)

Update Frontend (esp32-ota):

```
cd "D:\Automation Engineer\WebAutomation\WebESp" git add . git commit -m "pesan perubahan"
git push
```

Update Backend (local-server):

```
cd "D:\Automation Engineer\WebAutomation\local-server" git add . git commit -m "pesan
perubahan" git push
```

6. Git — PC Server (Ambil Update)

Pull update frontend:

```
cd /mnt/c/users/itd33/esp32-ota git pull
```

Pull update backend:

```
cd ~/local-server git pull # Restart uvicorn setelah pull
```

7. Struktur File

Backend (local-server):

```
local-server/ ■■■ main.py # Entry point FastAPI ■■■ routers/ ■■■ __init__.py # File kosong
■■■ ota.py # Semua endpoint ESP32 OTA
```

Frontend (esp32-ota):

```
esp32-ota/ ■■■ frontend/ ■■■ src/ ■■■ main.jsx # Entry point React ■■■
ESP32OTADashboard.jsx # Dashboard utama
```

Tambah project baru di backend:

```
# 1. Buat file baru routers/project_baru.py # 2. Tambah di main.py from routers import
project_baru app.include_router(project_baru.router, prefix="/project_baru")
```

8. Kode ESP32 — Kirim Telemetry Setelah OTA

Tambahkan kode ini di ESP32 setelah OTA selesai:

```
#include <HTTPClient.h> void sendTelemetry() { HTTPClient http;
http.begin("http://192.168.12.214:8000/ota/device/status"); http.addHeader("Content-Type",
"application/json"); float battery = analogRead(34) * (4.2 / 4095.0); // sesuaikan pin float
temp = temperatureRead(); // internal sensor int rssi = WiFi.RSSI(); bool saved = true;
String body = "{\"battery\":" + String(battery, 2) + ",\"temperature\":" + String(temp, 1) +
```

```
",\"rssi\":" + String(rssi) + ",\"save_to_ota\":true}"; http.POST(body); http.end(); }
```

9. Troubleshooting

Masalah	Solusi
`API merah / tidak konek`	Pastikan uvicorn jalan di WSL dan port proxy aktif
`git push gagal 403`	Gunakan token GitHub di URL remote
`npm not found di WSL`	sudo apt install nodejs npm -y
`uvicorn not found`	python3 -m uvicorn (pakai python3 -m)
`Port 8000 tidak bisa diakses`	Jalankan netsh portproxy dan firewall rule
`Card 05 tidak tampil`	Pastikan ESP32 sudah POST ke /ota/device/status